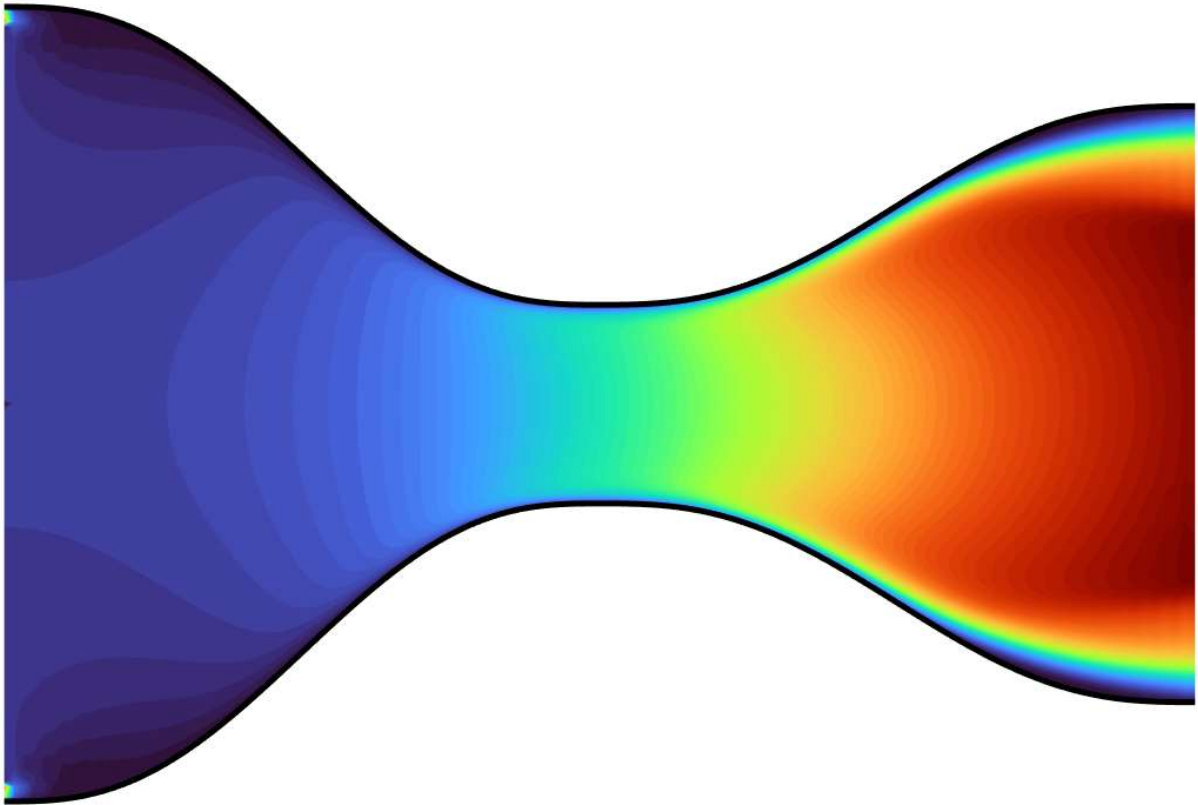


CFD Simulation of Viscous Compressible Flow in a Converging-Diverging Nozzle



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Abstract

A two-dimensional CFD solver was developed to simulate viscous compressible flow through a converging-diverging nozzle. The compressible Navier-Stokes equations were solved in conservative form using a structured finite-difference method with a MacCormack predictor-corrector scheme. A body-fitted coordinate transformation was used to map the curved nozzle geometry onto a rectangular computational domain.

The simulation captures key flow features, including subsonic acceleration in the converging section, choking at the throat, and supersonic expansion in the diverging region. Pressure, temperature, and density decrease along the nozzle, consistent with compressible flow theory. Boundary-layer development is observed along the wall due to viscous effects, while centerline results show strong agreement with quasi-one-dimensional isentropic predictions.

Overall, the solver produces physically consistent results and demonstrates the effectiveness of structured-grid methods for modeling compressible nozzle flows. Potential improvements include enhanced boundary condition treatment and extension to axisymmetric or three-dimensional configurations.

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Nomenclature

Symbol	Description	Units
ρ	Density	[-]
u	Velocity in x-direction	[-]
v	Velocity in y-direction	[-]
p	Pressure	[-]
T	Temperature	[-]
E	Total energy per unit mass	[-]
a	Speed of sound	[-]
γ	Ratio of specific heats	[-]
μ	Dynamic viscosity	[-]
k	Thermal conductivity	[-]
q_x, q_y	Heat flux components	[-]
$\tau_{xx}, \tau_{yy}, \tau_{xy}$	Viscous stress tensor components	[-]
U	Vector of conserved variables	[-]
F, G	Inviscid flux vectors (x, y directions)	[-]
V_1, V_2	Viscous flux vectors (x, y directions)	[-]
x, y	Physical coordinates	[-]
ξ, η	Computational coordinates	[-]
J	Jacobian of transformation	[-]
$h(\xi)$	Nozzle half-height function	[-]
L	Nozzle length	[-]
Δt	Time step	[-]
CFL	Courant-Friedrichs-Lewy number	[-]
D	Artificial dissipation term	[-]
$\epsilon_\xi, \epsilon_\eta$	Artificial dissipation coefficients	[-]
M	Mach number	[-]
p_t	Stagnation pressure	[-]
T_t	Stagnation temperature	[-]

1 Introduction

The objective of this project is to simulate compressible flow through a two-dimensional converging-diverging nozzle using a finite-difference computational fluid dynamics solver. The nozzle is comprised of a smooth converging section, a minimum-area throat, then a diverging section downstream of the throat. This geometry captures the key physics of compressible nozzle flow, the fluid accelerates in the converging section, choking at the throat, and continues to accelerate to supersonic speeds in the diverging section.

This problem is modeled on a half-nozzle domain. The lower boundary is treated as a symmetry line representing the nozzle centerline. The upper boundary is the physical nozzle wall. This choice reduces computational cost while preserving the physics of the full symmetric geometry. The flow is assumed to be compressible and viscous, with a stationary adiabatic wall and prescribed inlet stagnation conditions. A back-pressure condition is applied to the outlet to control the downstream operating state.

Because the nozzle geometry is curved, the physical domain is mapped to a rectangular computational domain. This allows the governing equations to be solved on a structured grid while still representing the nozzle wall accurately. The solver is advanced in time until a steady state solution is reached, then the numerical solution is represented through contour plots and centerline profiles. These results are compared qualitatively against quasi-one-dimensional compressible nozzle theory.

2 Governing Equations

The equations that govern the physics of this problem are the compressible Navier-Stokes equations written in conservative form. Equations (2.1) - (2.4) represent the two-dimensional equivalent form.

$$\text{Continuity:} \quad \frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0 \quad (2.1)$$

$$\text{X-momentum:} \quad \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} = \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \quad (2.2)$$

$$\text{Y-momentum:} \quad \frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2 + p)}{\partial y} = \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \quad (2.3)$$

$$\text{Energy:} \quad \frac{\partial(\rho E)}{\partial t} + \frac{\partial(u(\rho E + p))}{\partial x} + \frac{\partial(v(\rho E + p))}{\partial y} = \frac{\partial}{\partial x}(u\tau_{xx} + v\tau_{xy} - q_x) + \frac{\partial}{\partial y}(u\tau_{xy} + v\tau_{yy} - q_y) \quad (2.4)$$

All equations are written in non-dimensional form. Several assumptions are used to reduce the physical problem to the numerical model. First, the flow is treated as two-dimensional. Second, gas is modeled as calorically perfect with constant specific heat ratio, meaning the equation of state given in equation (2.5) can be used in the solver and equation (2.6) can be used to relate total energy with pressure and velocity. The speed of sound is calculated with equation (2.7).

$$p = \rho T \quad (2.5)$$

$$E = \frac{p}{\rho(\gamma-1)} + \frac{u^2 + v^2}{2} \quad (2.6)$$

$$a = \sqrt{\gamma T} \quad (2.7)$$

The third assumption takes viscosity and thermal conductivity as constant properties rather than temperature dependent. Fourth, the nozzle wall is stationary, no-slip, and adiabatic. The fifth assumption is that gravity is neglected, which is reflected in the Navier-Stokes equations above. The viscous terms are based on a Newtonian fluid model with Fourier heat conduction.

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (2.8)$$

$$\tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3}\mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad (2.9)$$

$$\tau_{xy} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) \quad (2.10)$$

$$q_x = -k \frac{\partial T}{\partial x} \quad (2.11)$$

$$q_y = -k \frac{\partial T}{\partial y} \quad (2.12)$$

For numerical modeling, the governing equations are written compactly in vector form as

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y}, \quad (2.13)$$

where

$$U = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix} \quad F = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix} \quad G = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix} \quad V_1 = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix} \quad V_2 = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \quad (2.14)$$

3 Coordinate Transformation

3.1 General Coordinate Transformation

Because the nozzle wall is curved, the physical flow domain is not rectangular in Cartesian coordinates. To apply a structured finite-difference method effectively, the physical domain is mapped to a rectangular computational domain defined by the coordinates (ξ, η) . The transformation is defined by

$$x = \xi, \quad y = \eta h(\xi). \quad (3.1.1)$$

Where $h(\xi)$ is the function of the nozzle height. The streamwise coordinate remains unchanged while the transverse coordinate is scaled by the nozzle height. The lower boundary $\eta=0$ represents the nozzle centerline while the upper boundary $\eta=1$ corresponds to the nozzle wall.

The Jacobian of the transformation is

$$J = \frac{\partial(x, y)}{\partial(\xi, \eta)} = h(\xi). \quad (3.1.2)$$

Which represents the local stretching between the computational and physical domains. Applying the chain rule gives the derivative relations

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} - \left(\frac{\eta \xi}{h} \right) \frac{\partial}{\partial \eta}, \quad (3.1.3)$$

$$\frac{\partial}{\partial y} = \frac{1}{h} \frac{\partial}{\partial \eta}. \quad (3.1.4)$$

Which are used throughout the solver to convert fluxes between physical and computational space.

3.2 Transformation of Inviscid Fluxes

The compressible Navier-Stokes equations are first written in conservative Cartesian form as shown in equation (2.13). The inviscid flux vectors F and G correspond to the x and y directions. After the coordinate transformation, these fluxes must be rewritten in terms of computational coordinates.

Using the mapping relations, the transformed inviscid fluxes become

$$\hat{F} = hF, \quad \hat{G} = G - \eta h_\xi F. \quad (3.2.1)$$

These transformed fluxes allow the governing equations to be discretized directly on the rectangular computational mesh.

3.3 Transformation of Viscous Fluxes

The viscous terms require physical derivatives of velocity and temperature. Since the solver stores the solution on the computational grid, these derivatives are first computed in computational space using finite differences. These computational derivatives are converted to physical derivatives using the transformation metrics. Applying the chain rule gives

$$u_x = u_\xi - \frac{\eta h_\xi}{h} u_\eta, \quad (3.3.1)$$

$$u_y = \frac{1}{h} u_\eta, \quad (3.3.2)$$

$$v_x = v_\xi - \frac{\eta h_\xi}{h} v_\eta, \quad (3.3.3)$$

$$v_y = \frac{1}{h} v_\eta, \quad (3.3.4)$$

$$T_x = T_\xi - \frac{\eta h_\xi}{h} T_\eta, \quad (3.3.5)$$

$$T_y = \frac{1}{h} T_\eta. \quad (3.3.6)$$

These physical derivatives are used to form the viscous stress tensor and heat flux vector. The viscous fluxes are first assembled in physical Cartesian form and are then transformed into computational space in the same way as the inviscid terms:

$$V = hV_1, \quad (3.3.7)$$

$$W = V_2 - \eta h_\xi V_1. \quad (3.3.8)$$

Here, V_1 and V_2 are the physical viscous flux vectors in the x and y directions and the vectors V and W are the transformed vectors in the computational domain.

4 Nozzle Geometry and Grid Definition

The nozzle geometry used in this project is a smooth two-dimensional converging-diverging nozzle with a nondimensional length of $L=3$, inlet height of 1, throat height of 0.25, and exit height of 0.75. The flow is modeled on a half-nozzle domain with a symmetrical boundary to accurately represent a converging-diverging nozzle. This gives the nozzle an exit-to-throat area ratio of 3, which is sufficient to support supersonic expansion with the correct back pressure.

To correctly model a wall shape that transitions smoothly between the converging and diverging section, a quintic blending function is used. This function ensures a smooth transition with continuous slope at both ends of the interval, avoiding geometric discontinuities that could create numerical difficulties in the metric terms and in the flow solution.

$$\begin{cases} h(x) = 1 - 0.75 \left[10 \left(\frac{x}{1.5} \right)^3 - 15 \left(\frac{x}{1.5} \right)^4 + 6 \left(\frac{x}{1.5} \right)^5 \right] & 0 \leq x \leq 1.5 \\ h(x) = 0.25 + 0.5 \left[10 \left(\frac{x-1.5}{1.5} \right)^3 - 15 \left(\frac{x-1.5}{1.5} \right)^4 + 6 \left(\frac{x-1.5}{1.5} \right)^5 \right] & 1.5 \leq x \leq 3 \end{cases} \quad (4.1)$$

Once the geometry is defined, a structured computational grid is generated in the rectangular computational domain (ξ, η). The streamwise coordinate is distributed uniformly from 0 to L, and the transverse coordinate is distributed uniformly from 0 to 1. These coordinates are then mapped into the physical nozzle using equation (3.1.1).

Figure 4.1 shows the final mapped grid used in the simulation. The grid remains logically rectangular in computational space but conforms smoothly to the converging-diverging nozzle geometry in physical space.

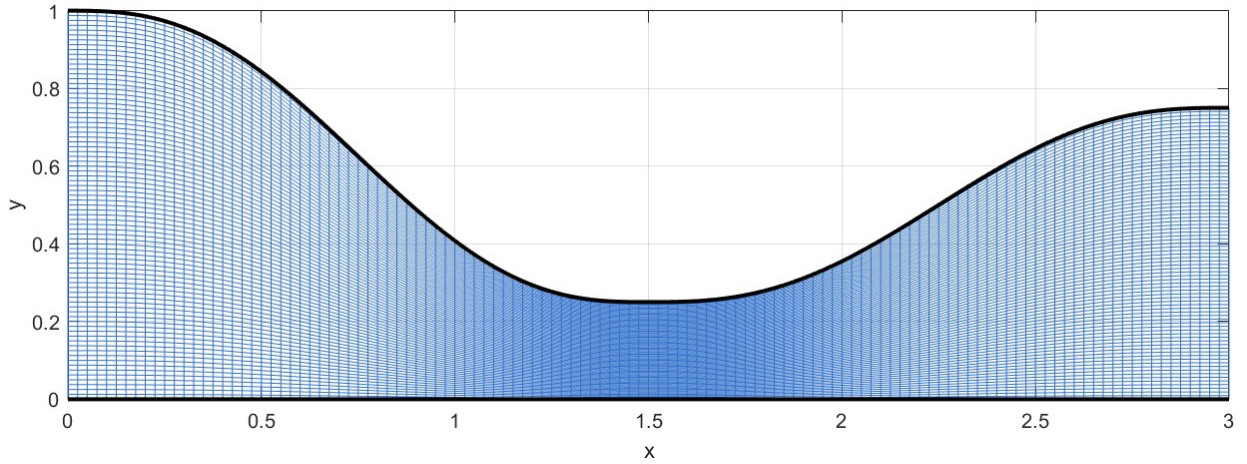


Figure 4.1: Structured Computational Grid Mapped to Physical Nozzle Domain

5 Numerical Method

5.1 Discretization Strategy

The governing equations are discretized in conservative form on a uniform computational mesh. The transformed inviscid and viscous fluxes are evaluated at each grid point, and their spatial derivatives are approximated using finite differences in the ξ and η directions. The solution U , containing the conserved variables given in equation (2.1.4), is advanced explicitly in time using a two-step MacCormack predictor-corrector scheme.

5.2 MacCormack Predictor-Corrector Scheme

The time integration is performed using a MacCormack predictor-corrector method. The predictor step uses forward difference schemes to evaluate the transformed flux derivatives. This produces the predicted state U^* in equation (5.2.1).

$$U_{i,j}^* = U_{i,j}^n + \frac{\Delta t}{J_{i,j}} \left[-\frac{\hat{F}_{i+1,j}^n - \hat{F}_{i,j}^n}{\Delta \xi} - \frac{\hat{G}_{i,j+1}^n - \hat{G}_{i,j}^n}{\Delta \eta} + \frac{V_{i+1,j}^n - V_{i,j}^n}{\Delta \xi} + \frac{W_{i,j+1}^n - W_{i,j}^n}{\Delta \eta} + D_{i,j}^n \right] \quad (5.2.1)$$

The fluxes are rebuilt from the predicted state U^* to form the predicted fluxes. In the corrector step, backward differences are applied to the predicted state and fluxes, and the final solution at the new time level is obtained using Equation (5.2.2).

$$U_{i,j}^{n+1} = \frac{1}{2} \left[U_{i,j}^n + U_{i,j}^* + \frac{\Delta t}{J_{i,j}} \left(-\frac{\hat{F}_{i,j}^* - \hat{F}_{i-1,j}^*}{\Delta \xi} - \frac{\hat{G}_{i,j}^* - \hat{G}_{i,j-1}^*}{\Delta \eta} + \frac{V_{i,j}^* - V_{i-1,j}^*}{\Delta \xi} + \frac{W_{i,j}^* - W_{i,j-1}^*}{\Delta \eta} + D_{i,j}^* \right) \right] \quad (5.2.2)$$

This scheme provides second-order accuracy for smooth solutions and reduces directional bias by combining forward and backward difference schemes. Both the inviscid and viscous transformed fluxes are included in each stage along with artificial dissipation term D. Boundary conditions are applied after both the predictor and corrector steps so that the predicted and corrected states remain physically admissible.

5.3 Inviscid and Viscous Flux Evaluation

At each time step, the primitive variables ρ , u , v , and E are recovered from the conserved variables in the vector U before the fluxes are assembled. Pressure is obtained from the equation of state. Temperature and speed of sound are evaluated from the thermodynamic relations.

The inviscid fluxes are formed in Cartesian space before being transformed into computational space using the relations developed in Section 3. The viscous fluxes are calculated by computing velocity and temperature derivatives in computational space, then converting them into physical derivatives using the transformation metrics. These physical derivatives are used to construct the viscous stress tensor and heat flux vector, then the viscous fluxes are also transformed into computational space.

5.4 Time-Step Selection and Stability

Because the solver is explicit, the time step is restricted by stability conditions associated with compressive wave propagation and viscous diffusion. The convective-acoustic time-step restriction is given in equation (5.4.1). The wave speeds in the ξ and η directions are determined from the local flow velocity, the speed of sound, and the coordinate transformation metrics.

$$\Delta t_{\text{conv}} = CFL \div \left(\frac{|u| + a}{\Delta \xi} + \frac{|\eta_x u + \eta_y v| + a \sqrt{\eta_x^2 + \eta_y^2}}{\Delta \eta} \right) \quad (5.4.1)$$

The second restriction is imposed by viscous diffusion, which is dictated by equation (5.4.2) and is scaled by the square of the grid spacing.

$$\Delta t_{\text{visc}} = CFL \div \left[2\mu \left(\frac{1}{\Delta \xi^2} + \frac{1}{\Delta \eta^2} \right) \right] \quad (5.4.2)$$

The final time step used in each iteration of the solver is determined by the minimum value of the convective and viscous limits over the full domain, as described in equation (5.4.3). This ensures stability of the explicit update throughout the nozzle.

$$\Delta t = \min(\min(\Delta t_{\text{conv}}, \Delta t_{\text{visc}})) \quad (5.4.3)$$

5.5 Artificial Dissipation

To improve numerical stability and suppress nonphysical oscillations, a second-order artificial dissipation term is added to the transformed governing equations. This term acts as a small smoothing operator in both computational directions. In the ξ direction, the dissipation is written as

$$D_\xi \sim \epsilon_\xi \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{\Delta \xi^2} \quad (5.5.1)$$

and in the η direction as

$$D_\eta \sim \epsilon_\eta \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{\Delta\eta^2}. \quad (5.5.2)$$

Therefore, the total artificial dissipation added at each grid point is

$$D_{i,j} = \epsilon_\xi \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{\Delta\xi^2} + \epsilon_\eta \frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{\Delta\eta^2}. \quad (5.5.3)$$

Here, ϵ_ξ and ϵ_η blank control the amount of numerical smoothing in the streamwise and transverse computational directions. They do not represent physical viscosity. They are used purely for numerical stabilization. Their purpose is to damp spurious high-frequency oscillations while remaining small enough to avoid excessive smearing of the physical solution.

5.6 Solution Monitoring

The numerical solution is monitored throughout the simulation using numerical and physical indicators. After the predictor and corrector updates, the solution is checked for nonphysical values because explicit compressible-flow solvers can become unstable if negative density or pressure values appear during the update. To avoid this, the solver verifies that all conserved variables remain finite and that density and pressure remain positive throughout the computational domain.

In addition to the physical checks, the residual is computed at each iteration to monitor convergence. The residual is defined as the maximum absolute change in the conserved variables between consecutive time levels.

$$res = \max|U^{n+1} - U^n| \quad (5.6.1)$$

This provides a measure of how much the solution is changing from one step to the next. As the solution approaches steady state, the residual is expected to decrease and eventually level off.

Physically meaningful quantities are also tracked during the simulation. One of these is the throat Mach number, which is monitored at a grid point near the nozzle centerline and throat location. Tracking the throat Mach number provides an indication of whether the solution is approaching the expected operating regime. In a classic start-up nozzle flow, the throat Mach number is expected to approach unity as the flow chokes, while the fluid continues to accelerate in the diverging section.

6 Initial Conditions and Boundary Conditions

6.1 Initial Condition

The solution is initialized using a quasi-one-dimensional isentropic guess rather than a uniform flow field. This reduces the startup transient time and provides an initial state that is already consistent with compressible nozzle flow. The initial flow is subsonic throughout the nozzle, the Mach-number distribution is prescribed as a function of streamwise position using the area-Mach relation given in equation (6.1.1).

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}} \quad (6.1.1)$$

Using the subsonic branch of this relation, an initial Mach distribution $M(x)$ is generated from the local nozzle area. Once the Mach number is known, temperature and pressure are computed using the isentropic relations given in equations (6.1.2) and (6.1.3).

$$\frac{T_{t0}}{T} = 1 + \frac{\gamma - 1}{2} M^2 \quad (6.1.2)$$

$$\frac{p_{t0}}{p} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\gamma/(\gamma - 1)} \quad (6.1.3)$$

Density and total energy are computed from the thermodynamic relationships given by equations (2.5), (2.5), and (2.6). The velocity components are prescribed by

$$u = M\sqrt{\gamma T}, \quad v = 0. \quad (6.1.4)$$

This produces a smooth, subsonic state so that choking develops as the simulation progresses.

6.2 Inlet Boundary Condition

The inlet is modeled as a subsonic stagnation inlet. The total pressure and total temperature are prescribed, acting as a large upstream reservoir feeding the nozzle. Because the inflow is subsonic, one characteristic must be obtained from the interior flow rather than being specified independently. The characteristic quantity used is

$$J^- = u - \frac{2a}{\gamma - 1} \quad (6.2.1)$$

where u and a are taken from the first interior column. This characteristic is combined with the prescribed stagnation conditions to reconstruct the inlet static state. The total temperature relation is written as

$$T_t = T + \frac{\gamma - 1}{2\gamma}(u^2 + v^2). \quad (6.2.2)$$

Using thermodynamic relations, this allows the inlet velocity, temperature, pressure, density, and total energy to be recovered. The transverse inlet velocity is prescribed as zero.

6.3 Outlet Boundary Condition

The outlet is modeled using a relaxed back-pressure condition. The back pressure is chosen low enough to support supersonic expansion in the diverging section for an exit-to-throat area ratio of 3. This area ratio can be used with the supersonic branch of equation (6.1.1) to calculate the exit Mach number corresponding to supersonic expansion. That exit Mach number can then be substituted into Equation (6.1.3) to determine the pressure ratio associated with an ideally expanded supersonic exit flow. In the solver, the outlet pressure is updated using a relaxation between the final interior pressure and the prescribed back pressure:

$$p_{exit} = \alpha p_{int} + (1 - \alpha)p_{back} \quad (6.3.1)$$

where α is the relaxation factor. This reduces abrupt forcing at the outlet and improves numerical stability. The density and velocity are extrapolated from the final interior column using

$$\rho_{exit} = \rho_{int}, \quad u_{exit} = u_{int}, \quad v_{exit} = v_{int}. \quad (6.3.2)$$

The total energy is computed using the updated outlet pressure with

$$E_{exit} = \frac{p_{exit}}{(\gamma - 1)\rho_{exit}} + \frac{1}{2}(u_{exit}^2 + v_{exit}^2). \quad (6.3.3)$$

This boundary condition provides a stable way to impose the necessary back pressure for supersonic expansion while allowing the interior flow to evolve consistently.

6.4 Centerline Boundary Condition

Only half of the nozzle is simulated, so the lower boundary is treated as a symmetry line representing the nozzle centerline. Along a symmetry boundary, the normal velocity must vanish, and the scalar variables have zero normal gradient. In conservative form, density, streamwise momentum, and total energy are copied from the first interior row to enforce zero gradient, and the transverse momentum is reflected:

$$\rho_{1,i} = \rho_{2,i} \quad (6.4.1)$$

$$(\rho u)_{1,i} = (\rho u)_{2,i} \quad (6.4.2)$$

$$(\rho v)_{1,i} = -(\rho v)_{2,i} \quad (6.4.3)$$

$$(\rho E)_{1,i} = (\rho E)_{2,i} \quad (6.4.4)$$

This treatment enforces zero normal flow at the centerline while preserving symmetry of the overall solution.

6.5 Nozzle Wall Boundary Condition

The upper boundary is the physical nozzle wall which is modeled as a stationary no-slip, adiabatic wall. The no-slip condition is enforced by

$$u_{wall} = 0, \quad v_{wall} = 0. \quad (6.5.1)$$

The adiabatic condition is enforced by imposing zero normal temperature gradient at the wall using

$$T_{wall} = T_{int}. \quad (6.5.2)$$

The wall pressure is extrapolated from the interior using

$$p_{wall} = p_{int}. \quad (6.5.3)$$

Wall density is recovered using the equation of state and the total energy of the wall is computed from the thermodynamic state and zero wall velocity. This boundary condition produces the viscous low-speed region that develops along the upper wall.

6.6 Corner Treatment

The corner cells are treated individually because multiple boundary conditions overlap at these points. For example, at the inlet corners, the stagnation inlet condition overlaps with either the centerline symmetry condition or the nozzle wall boundary condition. This can cause spurious Mach-number spikes to develop in the corner regions.

To minimize these spikes, the inlet stagnation condition is applied only to the interior inlet rows, and the corner cells are assigned separately using values consistent with the adjacent symmetry or wall boundary. Additional cleanup is applied to the left corners to suppress artificial momentum spikes. Although small spikes still occur near the inlet-wall junction, the bulk flow field is protected.

7 Isentropic Reference Solution

To provide a reference for the CFD results, a quasi-one-dimensional isentropic nozzle solution is considered. This reference solution is based on inviscid compressible-flow theory and is used to estimate the ideal exit conditions for the nozzle geometry. This provides a useful baseline for comparison with the CFD solution.

The exit-to-throat area ratio of the nozzle is 3. Using the supersonic branch of the area-Mach relation,

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}, \quad (7.1)$$

the exit Mach number corresponding to supersonic expansion is found to be 2.63742. Once the exit Mach number is known, the ideal exit state can be determined using standard compressible-flow relations:

$$\frac{T_e}{T_{t0}} = \frac{1}{1 + \frac{\gamma-1}{2} M_e^2} = 0.4182, \quad (7.2)$$

$$\frac{p_e}{p_{t0}} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\gamma/(\gamma-1)} = 0.0473, \quad (7.3)$$

and

$$\frac{\rho_e}{\rho_{t0}} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-1/(\gamma-1)} = 0.1131. \quad (7.4)$$

Using the solver's nondimensional inlet stagnation state of $T_{t0}=1.0$ and $p_{t0}=2.11424$, the ideal exit conditions are

$$T_e = 0.4182, \quad p_e \approx 0.100, \quad \rho_e \approx 0.239. \quad (7.5)$$

This isentropic solution serves only as a reference and not as an exact match to the CFD results. The CFD model is two-dimensional and viscous. Deviations due to wall effects, boundary-layer development, and numerical diffusion are expected. However, the isentropic solution provides a useful benchmark for validating the flow behavior.

8 Results and Discussion

The simulation was performed on a 121x81 structured grid using a CFL number of 0.1 and advanced to $t = 10$ to reach steady state. The resulting flow field is shown in Figure 8.1, with corresponding centerline profiles in Figure 8.2.

8.1 Flow Field Contours

Figure 8.1 shows contours of Mach number, pressure, temperature, and density throughout the nozzle. The flow remains subsonic in the converging section and accelerates toward the throat, where it reaches approximately Mach 1 along the centerline. Downstream of the throat, the flow continues to accelerate in the diverging section, reaching fully supersonic conditions. This behavior is consistent with compressible nozzle flow theory.

The pressure, temperature, and density contours show a consistent decrease in the streamwise direction. This behavior agrees with isentropic expansion, where fluid acceleration is accompanied by reductions in thermodynamic properties.

The Mach and temperature plots show a developing boundary layer along the upper wall. This is caused by viscous effects. The no-slip condition causes lower Mach numbers near the wall compared to the core flow. Viscous dissipation and thermal diffusion contribute to the thermal boundary layer.

A small irregularity is present near the top left corner of the domain. This feature is likely caused by inconsistencies in the boundary condition treatment where the inlet and wall conditions intersect. While it does not significantly affect the overall solution, it highlights the sensitivity of compressible solvers to corner boundary implementations.

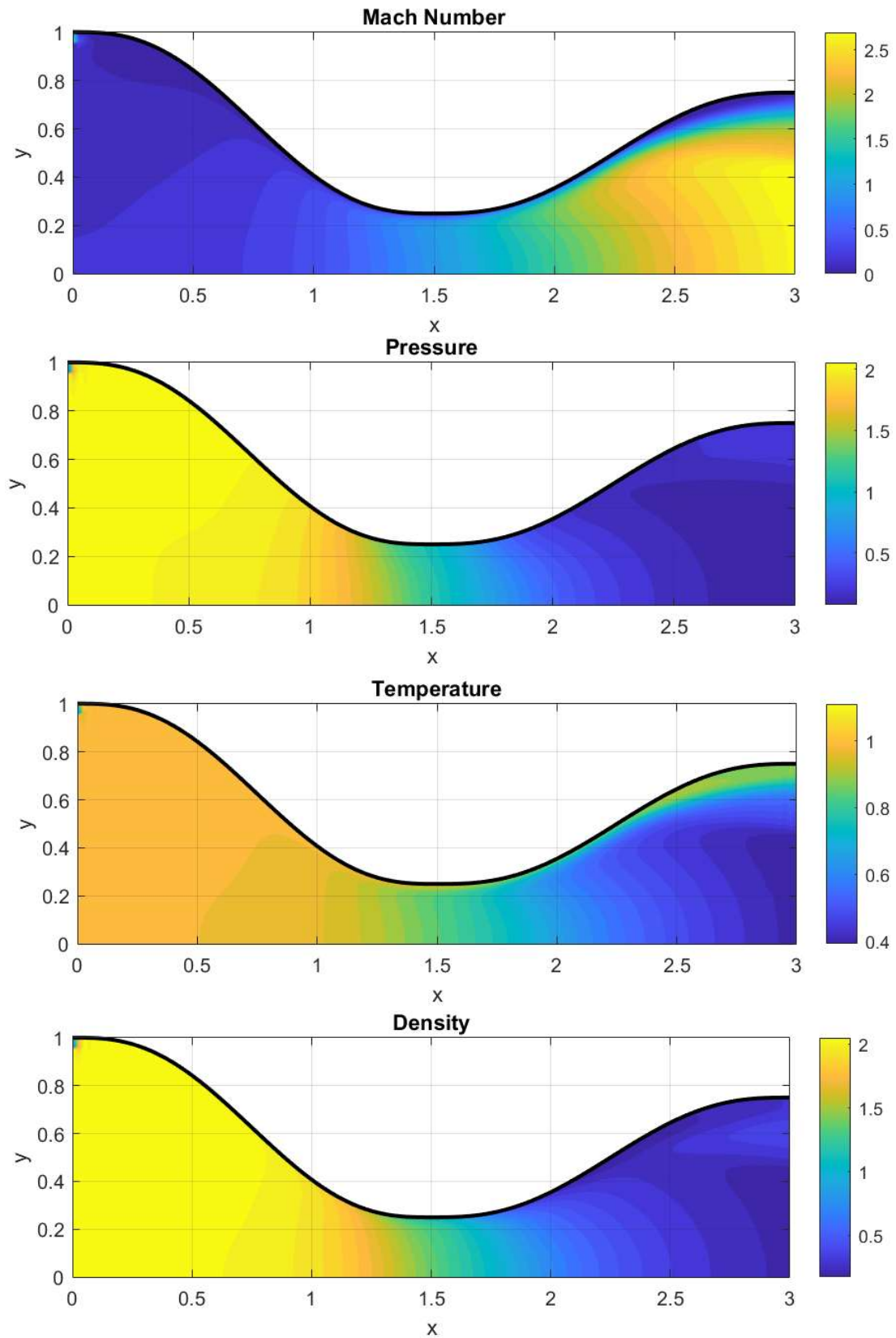


Figure 8.1: Contour plots of Mach Number, Pressure, Temperature, and Density at a steady state solution

8.2 Centerline Profiles

Figure 8.2 depicts the centerline distributions of Mach number, pressure, density, and temperature with the throat location indicated.

The Mach number increases smoothly from subsonic conditions at the inlet to approximately 1 at the throat, then continues to increase in the diverging section, reaching values around 2.6 near the exit. This behavior closely matches the isentropic reference solution.

The pressure, density, and temperature all decrease along the centerline, with increasingly rapid decreases downstream the throat. These trends are consistent with compressible flow acceleration and match the expected isentropic relationships along the centerline, where viscous effects are less dominant.

The agreement between the centerline solution and isentropic theory indicates that the solver is accurately capturing the dominant flow physics in the core region of the nozzle. The absence of a normal shock indicates that the imposed back pressure is sufficiently low to maintain fully supersonic flow throughout the diverging section.

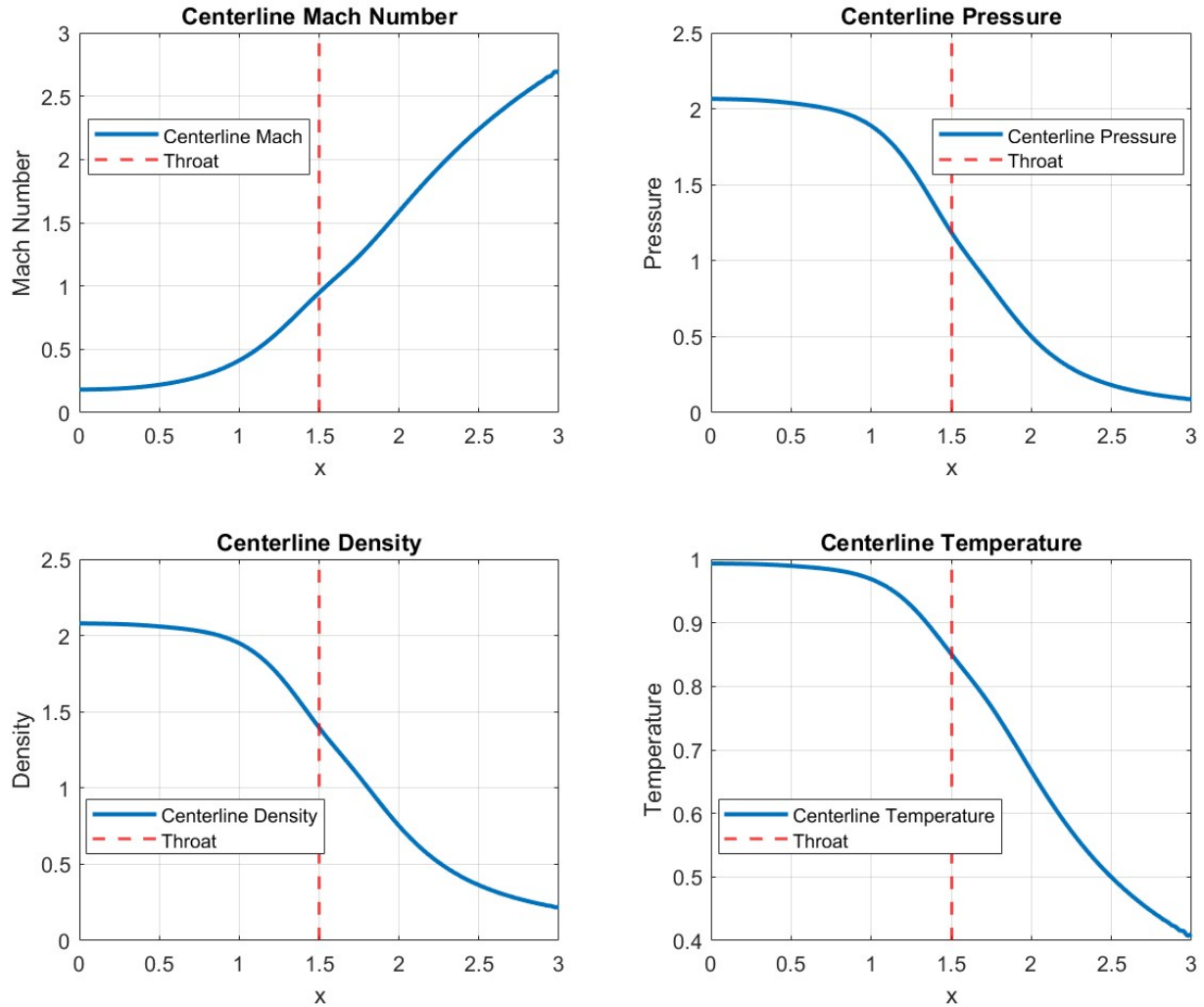


Figure 8.2: Centerline profiles of Mach Number, Pressure, Temperature, and Density at a steady state solution

8.3 Discussion of Physical and Numerical Effects

The results demonstrate that the solver successfully captures the key physical features of compressible nozzle flow. The flow correctly chokes at the throat and experiences supersonic expansion in the diverging section. The presence of boundary-layer effects near the wall confirms that viscous terms are being properly resolved.

The minor numerical artifacts observed near the boundary intersections are likely due to the simplified treatment of corner boundary conditions and the use of artificial dissipation.

Overall, the solution exhibits physically consistent behavior and agreement with theoretical expectations. This indicates that the numerical method, grid transformation, and boundary conditions are functioning as intended.

9 Conclusion

In this project, a two-dimensional CFD solver was developed to simulate viscous compressible flow through a converging-diverging nozzle. The compressible Navier-Stokes equations were solved in conservative form using a structured finite-difference approach with a MacCormack predictor-corrector scheme. The curved nozzle geometry was mapped onto a rectangular computational domain using a body-fitted coordinate transformation, enabling efficient numerical implementation.

The results show that the solver successfully captures the key features of compressible nozzle flow. The flow accelerates through the converging section, reaches sonic conditions at the throat, and continues to accelerate to supersonic speeds in the diverging section. The pressure, temperature, and density decrease in the streamwise direction, consistent with compressible flow theory. Boundary-layer development along the wall confirms that viscous effects are being resolved. The centerline profile shows strong agreement with quasi-one-dimensional isentropic behavior.

Minor numerical inconsistencies were observed near boundary intersections, particularly at the inlet-wall corner, indicating sensitivity to boundary condition treatment. Additionally, artificial dissipation, while necessary for stability, introduces slight numerical diffusion.

Overall, the solver produces physically consistent results that align well with theoretical expectations, demonstrating that the numerical method, coordinate transformation, and boundary conditions are functioning effectively.

Future work may focus on improving boundary condition robustness, refining artificial dissipation, and performing grid refinement studies to assess convergence. The solver could also be extended to three dimensions through an axisymmetric formulation, allowing more realistic modeling of nozzle flows while retaining computational efficiency.